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Chitosan nanoparticles as a smart nanocarrier for gefitinib for tackling lung cancer: Design of experiment and *in vitro* cytotoxicity study

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ABSTRACT

Due to its poor solubility and systemic side effects, gefitinib (Gef) has limited application in treatment of lung cancer. In this study, we used design of experiment (DOE) tools to gain the necessary knowledge for the synthesis of high-quality gefitinib loaded chitosan nanoparticles (Gef-CSNPs) capable of delivering and concentrating Gef at A549 cells, thereby increasing therapeutic effectiveness while decreasing adverse effects. The optimized Gef-CSNPs were characterized by SEM, TEM, DSC, XRD, and FTIR analyses. The optimized Gef-CSNPs had a particle size of 158±3.6 nm, an entrapment efficiency of 93±1.2 %, and a release of 97±0.6 % after 8 h. The *in vitro* cytotoxicity of the optimized Gef-CSNPs was found to be significantly higher than pure Gef ($IC_{50} = 10.08 \pm 0.76$ μg/mL and $IC_{50} = 21.65 \pm 0.32$ μg/mL), respectively. In the A549 human cell line, the optimized Gef-CSNPs formula outperformed pure Gef in terms of cellular uptake (3.286 ± 0.12 μg/mL and 1.777 ± 0.1 μg/mL) and apoptotic population (64.82 ± 1.25 % and 29.38 ± 1.11 %), respectively. These findings explain why researchers are so interested in using natural biopolymers to combat lung cancer, and they paint an optimistic picture of their potential as a promising tool in the fight against lung cancer.

1. Introduction

The statistics show how many people are diagnosed with cancer and how many die from it each year. According to the National Cancer Institute (NCI), the estimated number of new cancer cases in the United States in 2020 was 1,806,590, and the estimated number of cancer deaths was 606,520 during the same period [1]. Colorectal, lung, and prostate cancers accounted for approximately 49 % of all cancers in men in 2023, while colorectal, lung, and breast cancers accounted for approximately 52 % of all cancers in women [2].

Among all cancers, lung cancer has the highest mortality rate and the second highest incidence rate [3]. Lung cancer is a horrifying, devastating, and grueling disease The estimated number of new lung cancer cases in the United States in 2022 is 236,740 (12.3 % of all new cancer cases), with 130,180 deaths (21.4 % of all cancer mortality). Lung cancer kills more people (21.4 % of all cancer deaths) than colon (8.6 %), prostate (5.7 %), and breast (7.1 %) cancers combined [4]. While the statistics on lung cancer are concerning, they are also empowering. As a result, we set out to address and find out a solution to this stubborn killer "Lung cancer".

This effort is not designed to provide a lung cancer cure. That for now, remains impossible. But we add a new piece of information that gives us one more tool to add to our arsenal in our battle against cancer "The emperor of all maladies." We launched a new attempt to combat a disease that has impacted almost everyone's life by developing new weapons to tackle lung cancer.

Chemotherapy drugs are selective toxins that kill rapidly growing cells first. Chemotherapy drugs cause some level of harm to all cells in the body even healthy ones, but they have the most potent effect on rapidly dividing cells which is exactly the duty of cancer. Because cancer cells grow quickly, they are vulnerable to chemotherapy. If doctors are lucky, chemotherapy drugs will kill the cancer before killing the patient. Rapidly growing normal cells, like hair follicles, stomach and intestinal linings affected by chemo drugs as well, leading to the known adverse effects of baldness, nausea, and vomiting [5]. This lack of selectivity of chemo drugs added to their lack of efficacy making them

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underperformers in the battle against lung cancer. This energized the researchers to create a more selective toxin which is "targeted therapy" or precision-targeted chemotherapy (precision-guided weapons).

Targeted medications promised greater efficacy while being less toxic than chemotherapy. If chemotherapy is like carpet bombing, then targeted therapy is like the cancer arsenal's "smart bombs," hitting specific targets to destroy cancer cells while causing little collateral damage.

Gef is an example of the genetics-focused approach to cancer treatment. Gef specifically targets cells with EGFR gene mutations. The EGFR is a protein that resides on the surface of cells [6]. Its normal function is to drive cell division and growth. The EGFRs are abundant in NSCLC cells, which promote faster growth. EGFR inhibitors block the EGFR signal that tells cells to grow [7]. Gef is an orally administered drug that inhibits the tyrosine kinase domain of the EGFR. Gef is a first-line therapy for locally advanced or metastatic non-small cell lung cancer, was granted FDA approval in 2015 [8]. The use of Gef (targeted drug therapy) was thought to be an ultimate game changer in the treatment of lung cancer, but unfortunately its use was accompanied by tremendous downsides. Gef possesses a very poor aqueous solubility and low oral absorption. This poor water solubility, slow intestinal absorption, and low bioavailability compromise its therapeutic efficacy [9]. Furthermore, Gef has a large volume of distribution (1400 L) which will extend its effects far beyond its target site and lead to widespread distribution into normal tissues, resulting in horrifying adverse effects such as hepatotoxicity, rash or skin toxicity, diarrhea, and interstitial lung disease. So, instead of chasing lung cancer, we will chase the adverse effects of Gef.

Despite all the doom and gloom, there are some encouraging signs, as nanomedicine has advanced significantly over the last two decades of intensive research [10]. The term "nano" has grown in popularity not only among scientists but also among the general public [66]. The advancements in nanoparticle-based drug delivery systems, primarily for drug delivery to solid tumors, have been astounding. The rapid advancement in nanomedicine has provided reason for optimism. Nanoparticles work like a vehicle or a truck (a "nano-truck") to deliver the drug exclusively to its target organ [11]. We can load Gef inside the NPs and they will act as the carrier to transport Gef into the tumor's heart. The NPs can keep Gef from being destroyed within the body. In contrast to Gef therapy alone, NPs could identify bad cells while saving healthy ones. Thanks to the enhanced permeation and retention effects (EPR), the so-called passive targeting, which relies on the existence of fenestrations in the tumor blood vessels (in the endothelium). These fenestrations allow NPs to enter the cancer cells (enhanced permeability). Furthermore, the lack of lymphatic drainage in the tumor tissue makes recapture of these nanoparticles impossible (enhanced retention) [10,12].

In this work, we utilized one of the most important, versatile, and widely available biopolymer: chitosan to act as a nanocarrier for Gef. Chitosan has gotten a lot of attention among all the promising nanocarriers. It is a linear polysaccharide with excellent biocompatibility, abundance, biodegradability and mucoadhesion properties [12]. Because of its mucoadhesive properties, it prolongs Gef residence time at the lungs, resulting in a significant increase in bioavailability. This mucoadhesion property is the weapon that chitosan employs to interact with the negatively charged mucus [12] and thus extend its residence time. Furthermore, chitosan possesses marvelous anticancer, antimicrobial, immunoregulatory, and other biological properties. It has found widespread application in drug delivery and other biomedical applications. Sumatriptan [13], primaquine [14], aspirin [15], simvastatin [16], calcitonin [17], cisplatin [18], methotrexate [19], and other drugs have been delivered using chitosan nanoparticles (CSNPs). The most common method for producing CSNPs is the "ionic gelation technique" or "ionic cross-linking," which employs an anionic cross-linker, sodium tripolyphosphate (TPP). Chitosan, as a biopolymer, provides the NPs with a "biological identity" that aids cellular uptake and thus efficacy. Furthermore, assist them in overcoming the fundamental obstacle in cancer treatment, which is the limited uptake of cancer drugs.

A full factorial design was used as a strategy to improve Gef-CSNPs synthesis and predict the best formula [20,65]. Drug content, entrapment efficiency (EE%), particle size (PS), transmission electron microscopy (TEM), scanning electron microscopy (SEM), and *in vitro* release studies were all performed on the desired formula. To investigate the possible interaction between Gef and CSNPs, it was also subjected to physicochemical characterization using X-ray diffraction (XRD), differential scanning calorimetry (DSC), and Fourier transform infrared spectroscopy (FTIR).

In this study, we looked at the system's ability to deliver its cargo into A549 human lung cancer cells in order to evaluate the efficacy of Gef-CSNPs *in vitro*. The HPLC study showed that Gef-CSNPs were more efficiently taken up by cells than pure Gef. The MTT experiment exhibited substantially higher cytotoxicity of the optimized formula, while the flow cytometric assay showed much higher apoptosis of Gef-CSNPs compared to pure Gef.

2. Materials and methods

2.1. Materials

Sigma-Aldrich provided low molecular weight (LMW) chitosan with a degree of deacetylation of 75–85 % and an average MW of 50 KDa. Gefitinib was generously donated by King Saud University in Saudi Arabia. Tween® 80 (polyethylene sorbitan monooleate) and sodium tripolyphosphate (TPP) were supplied by Sigma-Aldrich Chemie GmbH. (Germany). Phosphate buffer saline was created using sodium hydrogen phosphate and potassium dihydrogen phosphate.

2.2. Preparation of Gef-CSNPs

Ionic gelation was used to create chitosan nanoparticles. Nanoparticles are formed through an ionic interaction between a positively charged chitosan solution and a negatively charged TPP solution [21].

To begin, chitosan solutions (2, 3, and 4 mg/mL, 50 mL) were made by dissolving chitosan in 1 % acetic acid. Following that, the chitosan solutions were treated with 100 mg Gef dissolved in 10 mL of water containing 1 % Tween® 80. The solutions were then magnetically stirred for 30 min at 800 rpm and 25 °C in 4 mL of water containing varying amounts of TPP (1, 2, and 3 mg/mL). Gef-CSNPs formed on their own. In the final stage, the nanoparticles were centrifuged for 30 min at 12000 rpm to isolate them. Finally, the supernatant was collected to determine the Gef encapsulation indirectly. For further research, the freeze-dried nanoparticle powders were kept in sealed containers.

Maltose, trehalose, and glycine were tested as diluents and lyoprotectants for freeze-drying nanoparticles. A microscope coupled to a lyophilization module, Lyostat 2, model FDCS 196 (Linkam Instruments, Surrey, UK), equipped with a liquid nitrogen freezing system (LNP94/2) and a programmable temperature controller, was used to determine the collapse temperatures of Gef-CSNPs without and with 3 % (*w/v*) lyoprotectants (TMS94, Linkam). A Pirani valve was used to monitor pressure. The equipment was calibrated using an aqueous NaCl solution (eutectic temperature of −21.1°C). Heating and cooling ramps of 5°*C*/min were applied to the samples. The freeze-drying conditions were the freezing temperature at − 70°C, the shelf-cooling rate at 2°C/*min*, and the primary drying temperature at − 40°C [22].

2.3. Experimental design

The design of experiment (DOE) was used as a statistical tool to investigate and comprehend the impact of the factors (alone and in combination) on the responses, allowing us to optimize the responses and analyze the factors studied [23].

A three-level, two-factor full factorial design (3^2) with JMP® software was used to investigate the effects of independent variables

(chitosan and TPP) on responses (PS, EE(%), and %release) (version 17, SAS, USA). To statistically explore the data, JMP® software analyzed it and visualized the results.

JMP® Software was used to create a 3^2-factorial design, analyze the results, statistically explore the data, and visually represent the findings. We also used predictive analytical tools, such as predictive profilers, to assist us in building, improving, developing, and predicting the model. The model specification was center polynomials, the personality was standard least squares, and the emphasis was effect screening. We used quality signs, which are multiple correlation coefficients (R^2) and adjusted R^2, to evaluate the model's fitness. A *p*-value of <0.05 is considered statistically significant, and we evaluated the model's fitness using *p*-values <0.05. The interaction and relationship between the independent variables (factors) chitosan and TPP and the dependent variables: PS (nm), EE (%), and %release were also shown.

As a result, we employed a full factorial design to model the desired responses and provide a thorough understanding of the interacting variables. The optimization step was carried out using the desirability function.

2.4. Entrapment efficiency (%)

The EE (%) of the various nanoformulations (F1–F9) was determined using the indirect method, which involved centrifuging freshly prepared nanoparticle dispersion at 12000 rpm for 30 min. The amount of free Gef in the supernatant was determined using UV spectroscopy at 250 nm (Shimadzu 1601; Shimadzu, Kyoto, Japan). The efficiency of Gef entrapment was determined by comparing it to a standard curve obtained from a Gef solution in methanol at concentrations ranging from 2.5 to 12.5 μg/mL. All analyses were performed in triplicate. The EE (%) was determined as follows [24]:

$$\text{EE}\ (\%) = \left(\frac{\text{Total Gef} - \text{Free Gef}}{\text{Total Gef}}\right) \times 100 \quad (1)$$

2.5. In vitro release study of Gef and Gef-CSNPs

A dialysis bag (12 KDa MW cutoff, India) containing the Gef-CSNPs (5 mg) was placed in a 50 mL FBS (pH 5) at 37°*C* with stirring rate (100 rpm) [25].

At predefined intervals (0.25, 0.5, 1, 2, 4, and 8 h), 4 mL samples were taken from the dissolution medium, and replaced with an equal volume of fresh buffer. The removed sample was spectrophotometrically examined at an absorbance of 250 nm (Shimadzu 1601; Shimadzu, Kyoto, Japan), and the findings were recorded [26]. The linear regression curve that was derived from the Gef standard curve was used to calculate the concentration of the released Gef. The %release of Gef was computed and expressed as the mean ±SD (n= 3).

2.6. Characterization

2.6.1. Fourier transform infrared (FT-IR) spectroscopy

Thermo Scientific Nicolet iS50 FTIR spectrometer, USA, captured the FTIR spectra of pure Gef, chitosan, TPP, plain CSNPs, and Gef-CSNPs. The ground-up, dry sample was combined with KBr before being crushed into a pellet. In the 4000 cm^{-1} to 400 cm^{-1} wavenumber region, FTIR spectra were obtained [27].

2.6.2. Particle size (PS) and morphological characterization by DLS, transmission electron microscopy (TEM), and scanning electron microscopy (SEM)

All batches were evaluated for PS (nm), polydispersity index (PDI), and zeta potential using the dynamic light scattering method (Zetasizer Nano Series Nano-ZS, Malvern Instruments, Malvern) at a 90° measurement angle and a temperature of 25°*C*.

The best formula's size and shape were examined using TEM (JEOL, JEM-2100, Japan). One drop of the diluted nanoparticle suspension was applied to a copper grid and left to dry and images were obtained at operating voltage of 100 kV. Plain CSNPs and Gef-CSNPs were examined using SEM (Quanta FEG-250 SEM instrument, USA) [28]. A small quantity of samples were placed in aluminum stubs before being coated with platinum. The SEM images were obtained using a 20 kV acceleration voltage [29].

2.6.3. Differential scanning calorimetry (DSC)

DSC is used to determine how Gef and CSNPs interact chemically. France's LABSYS evo Setaram carried out the DSC measurements. On aluminum pans, samples weighing 2–10 mg were sealed. The probes were exposed to temperatures between 25 and 250 °C at a rate of 10 °C per minute in a nitrogen atmosphere [30].

2.6.4. X-ray diffraction (XRD)

To ascertain the crystallinity of pure Gef, chitosan, and Gef-CSNPs, XRD analysis was carried out on a Bruker D8 Discover diffractometer outfitted with a Cu-Kα X-ray source (on lab scale X-ray instruments, copper Kα is a common X-ray energy) that emits radiation at a wavelength of 1.54 A° through a copper-sealed tube. The samples were examined at 2θ between 5 and 80°. And relative intensity data were gathered [31].

2.7. Cell viability assay

In the mitochondria of living cells, yellow MTT (3-(4, 5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, a tetrazole) is reduced to purple formazan. Because this reduction occurs only with active mitochondrial reductase enzymes, conversion is an important indicator of the number of viable cells.

Cytotoxicity was studied in a non-small cell lung cancer cell line (A549) (Nawah-Scientific, Cairo, Egypt). Cells were cultured in DMEM medium containing 10 % FBS, 100 units/mL penicillin, and 100 mg/mL streptomycin [32]. Cultures were kept at 37 °C in a humidified atmosphere with 5 % CO_2.

To make a stock solution, each sample was diluted in DMEM complete media at 37 °C.

There were ten concentrations of two-fold serial dilutions performed.

Confluent monolayers of cells were grown for 24 h in 96 well microtiter plates. Cells were incubated in triplicate with various concentrations of the test samples for 72 h at 37 °C in a CO_2 environment. After that, add 20 μL of 5 mg/mL MTT to each well and incubate for 4 h at 37 °C. The medium was then carefully removed, and 150 μL of MTT solvent was added. The cells were agitated on an orbital shaker for 15 min while covered in tinfoil. Finally, the optical density (OD) was measured in a microplate reader (BMGLABTECH® Fluostar Omega, Germany) at 570 nm.

2.8. Evaluation of apoptosis by Annexin-V/PI staining assay

An Annexin V-FITC apoptosis detection kit (Abcam Inc., Cambridge Science Park, Cambridge, UK) combined with 2-fluorescent channel flow cytometry is used to assess apoptotic pathways. After 72 h of treatment with test compounds and 48 h of treatment with doxorubicin (10 μM) as a positive control, cells (10^5 cells) are trypsinized and washed twice with ice-cold PBS (pH 7.4). The cells were then incubated in the dark for 30 min with 0.5 mL of Annexin V-FITC/PI solution. After staining, cells were injected into an ACEA NovocyteTM flow cytometer (ACEA Biosciences Inc., San Diego, CA, USA) and analyzed for fluorescent signals of FITC and PI [33] using FL1 and FL2 detectors (λex/em 488/530 nm for FITC and λex/em 535/617 nm for PI). 12,000 events were collected for each sample, and positive FITC and/or PI cells were quantified and calculated using ACEA NovoExpressTM software (ACEA Biosciences Inc., San Diego, CA, USA).

2.9. In vitro cellular uptake

Lung cancer of the non-small cell type A549 cell lines were grown in DMEM/RPMI-1640 media supplemented with 100 mg/mL streptomycin, 100 units/mL penicillin, and 10 % heat-inactivated fetal bovine serum at 37°*C* in a humidified, 5 % (*v*/v) CO_2 atmosphere. For 24 h, cells were seeded in a T25 flask (Greiner Bio-One, Germany). The cells were then treated with fresh media containing the drug concentration that had been predetermined. After the specified time of incubation, cells with a confluence of 80 % were harvested using trypsin/EDTA. The pellet was then obtained by using a cooling centrifuge.

HPLC analysis:

a) Standard preparation: A Gef stock solution in methanol was prepared, followed by serial dilution to obtain concentrations of (0.1, 0.2, 0.4, 0.6, 0.8, and 1 μg/mL), filtered through a 0.22 μm syringe filter, and 100 μL injected.
b) Sample preparation: 0.5 mL Acetonitrile was added to each sample, which was then vortexed for 1 min, sonicated for 30 min, centrifuged for 15 min at 15,000 rpm, filtered through a 0.22 μm nylon syringe filter, and 100 μL was injected.

3. Results and discussion

3.1. Experimental design for Gef-CSNPs optimization

Table 1 displays the results for Gef-CSNPs under different conditions. JMP® software received the data for the responses. The observed and predicted values of all dependent variables were in agreement with the adjusted Rsquare. To examine the effect of independent variables on the measured responses, the response-surface methodology was used. Statistical design software was used to create the design matrix, which consisted of nine experimental runs.

According to the results of the investigation into how chitosan and TPP affected the chosen responses (Table 2), chitosan significantly affects the PS, the EE (%), and the %release, (*P*-values of 0.0006, 0.0018, 0.0014, respectively). TPP has a statistically significant effect on PS (a *P*-value of 0.0036), EE (%) (a P-value of 0.0011), and %release (a P-value of 0.0008).

To compare how the factors affected the responses, we used the Pareto chart (supplementary materials). It can compare factors regardless of their units. The baseline used to assess the validity of the parameters is the blue line in the Pareto chart. Factors exceeding the baseline significantly affect the PS, the EE (%), and the %release. Both chitosan and TPP are major factors because they are exceeding the baseline [34].

The *P*-value is used to test the hypothesis, if P-value < 0.05, it indicates that the factors affect the responses. Therefore, since the *P*-values of all factors involved in this study are <0.05, the factors are considered to affect the PS, the EE (%), and %release.

Chitosan and TPP both had a linear model, with P-values of 0.00062 and 0.00076, respectively. Chitosan and TPP interaction was insignificant (P-value = 0.06279) as presented in Table 2.

Table 2
Pareto results and Parameter estimates of PS, %EE, and %release illustrate that both chitosan and TPP have significant effects on PS (nm), EE (%), and %release.

Factor	Response	P-value	Std error
TPP	PS (nm)	0.0036*	7.45
TPP	EE (%)	0.0011*	1.89
TPP	%release	0.0008*	1.53
Chitosan	PS (nm)	0.0006*	7.45
Chitosan	EE (%)	0.0018*	1.89
Chitosan	%release	0.0014*	1.53

Pareto results (*P*-values):
CS; 0.00062, TPP; 0.00076, and TPP*CS; 0.06279.
A p-value < 0.05 is statistically significant (*).

The PS (nm), EE (%), and %release results were fitted to the linear model generated by the full factorial design, and the model's adequacy was verified using an ANOVA test [35]. PS (nm), EE (%), and %release had *p* values of 0.0013, 0.0016, and 0.0012, respectively. It is clear that the outcomes were highly significant as shown in Table 3.

The coefficient of determination (R^2) in a regression analysis is the value that reveals how much of the factor (dependent variable) is expressed by the explanatory variables in that analysis. The R^2 values of the model developed for all the responses (PS, EE (%), and %release) are close to one. The closeness of R^2 to one indicates the ease of use and accuracy of the generated model.

The R^2 value for the PS (nm) was 0.947, which indicates that 94.7 % could be assigned to the factors and 5.3 % of the total variations couldn't be explained by the model. The R^2 value for the EE (%) was 0.944, which indicates that 94.4 % could be assigned to the factors and 5.6 % of the total variations couldn't be explained by the model. The R^2 value for the %release was 0.949, which indicates that 94.9 % could be assigned to the factors and 5.1 % of the total variations couldn't be explained by the model. From Table 3, the highest R^2 values were found with the % release, followed by the PS, and finally the EE (%) [36].

The adjusted R^2 value demonstrates the model's generalizability, within which the design fits. To put it another way, if the model was derived from the population rather than the sample, it would explain the total variance at the adjusted R^2 level [37]. In this investigation, the model R^2 values were shown to be greater than the adjusted R^2 values for all response variables. Lower adjusted R^2 values for all responses showed that there was a significant relationship reflected in the model [38].

The predicted R^2 value illustrates the closeness between the predicted results of the model and the observed results of the universe [39]. The predicted *versus* actual plots compare the results of the experimental values of the responses with those of the model's predicted values. The full factorial design showed higher correlation and better agreement between experimental and predicted results, confirming its higher predicted R^2 obtained values [40]. The actual PS *versus* predicted PS plot

Table 1
Full factorial design run matrix and the obtained responses: Chitosan and TPP are the studied factors, while the PS, the EE (%), and the %release are the responses.

Formula	Independent variables		Dependent variables (Responses)		
	TPP (mg/mL)	Chitosan (mg/mL)	PS (nm)	EE (%)	%release
F1	1	2	106	71.3	77.2
F2	1	3	161	58.2	68.3
F3	1	4	182	49.4	57.9
F4	2	2	145	83.2	85.8
F5	2	3	172	64.5	73.1
F6	2	4	246	53.8	62.6
F7	3	2	158	92.7	97.1
F8	3	3	201	87.1	92.2
F9	3	4	321	75.6	81.1

Table 3
Summary of fit and ANOVA analysis of full factorial design for PS (nm), EE (%), and %release.

Model response	Sum of squares	Mean of square	P value	Rsquare	Rsqaure Adj	RMSE
PS (nm)	30,052.42	10,017.50	0.0013*	0.947	0.916	18.24
EE (%)	1760.89	586.96	0.0016*	0.944	0.909	4.62
%release	1321.26	440.42	0.0012*	0.949	0.919	3.76

* A p-value < 0.05 is statistically significant.

has an R^2 of 0.95 and a *P*-value of 0.0013; the actual EE (%) *versus* predicted EE (%) plot has an R^2 of 0.94 and a P-value of 0.0016; and the actual %release *versus* predicted %release plot has an R^2 of 0.95 and a P-value of 0.0012. The higher predicted R^2 values of the responses revealed an excellent fit to the model.

In this study, the bias (%) or Error (%) calculation was used to compare the actual and predicted response variables of the formulations to figure out how reliable the design model made for Gef-CSNPs was. The following equation was used to figure out the percentage of bias of the predicted and actual values of all nanoformulations:

$$\text{Bias }(\%) = \frac{[\text{Predicted} - \text{Actual}]}{\text{Actual}} \times 100 \quad (2)$$

The fact that the bias percentage values are under 15 % shows that there is good agreement between the formulation's predicted and actual response variables [41].

The prediction formula for PS (nm) was defined as:

$$PS\ (nm) = 188 + 38.5*(TPP-2) + 56.67*(CS-3) + (TPP-2)*[(CS-3)*21.75] \quad (3)$$

Increasing TPP or chitosan will raise the PS, according to this equation (Eq. (3)). All factors (TPP and chitosan) have a significant effect on the PS (nm).

The prediction formula for the EE (%) was defined as:

$$EE\ (\%) = 70.64 + 12.75*(TPP-2) + -11.4*(CS-3) + (TPP-2)*[(CS-3)*1.2] \quad (4)$$

According to Eq. (4), entrapment efficiency increases by rising TPP concentration, where it decreases by increasing chitosan concentration. Table 4 shows the details of the predicted and observed entrapment efficiency.

The prediction formula for the %release was defined as:

$$\%release = 77.26 + 11.17*(TPP-2) + -9.75*(CS-3) + (TPP-2)*[(CS-3)*0.825] \quad (5)$$

Eq. (5) shows that while %release decreases as chitosan concentration increases, it increases when TPP increases. The observed and actual %release are shown in Table 4.

It was clear that the model was successful in predicting the PS (nm), EE (%), and the %release, respectively.

Minimize PS, maximize EE (%), and maximize %release were the optimization parameters. F7 (Chitosan = 2 mg/mL and TPP = 3 mg/mL) was determined to be the best formula with a desirability value of 0.887 (prediction profiler is provided in supplementary materials).

3.1.1. *The impact of the CS on PS, EE (%), and %release using a full factorial design*

At chitosan = 2 mg/mL, the mean PS (F1, F4, and F7) was 136 nm. While chitosan = 3 mg/mL increased the mean PS to 178 nm (F2, F5, and F8), chitosan = 4 mg/mL increased the mean PS to 250 nm (F3, F6, and F9) as presented in Fig. 1. The data clearly show that as the chitosan is raised, the PS rises. The most influential factor on PS was chitosan concentration, which had a *P*-value of 0.0006 compared to TPP's P-value of 0.0036 [42–44].

The particle size increased significantly as the chitosan concentration increased. It could be due to the increased availability of protonated amine groups for ionic gelation as chitosan concentration and viscosity increase [43,45].

The EE (%) ranges from 49.4 to 92.7 %, with a downward trend as chitosan increases. The mean EE (%) was 82 % at chitosan = 2 mg/mL (F1, F4, and F7), 70 % at chitosan = 3 mg/mL (F2, F5, and F8), and 60 % at chitosan = 4 mg/mL (F3, F6, and F9) as shown in Fig. 1. The data show that EE (%) decreases as chitosan increases.

Higher chitosan concentrations make the gelation medium highly viscous, preventing Gef entrapment. Lower chitosan viscosity at lower concentrations improves Gef entrapment and gelation between chitosan's positively charged amino groups and TPP's negatively charged groups [42].

Finally, chitosan had a significant impact on Gef release from CSNPs. The %release ranges from 57.9 to 97.1 %. Fig. 1 shows a downward trend as chitosan levels rise. Chitosan = 2 mg/mL (F1, F4, and F7) had a mean %release of 87 %, while chitosan = 3 mg/mL (F2, F5, and F8) had a mean %release of 78 %, and chitosan = 4 mg/mL (F3, F6, and F9) had a mean %release of 67 % as revealed in Fig. 1. The results show a clear picture of the contraction of the %release as chitosan levels rise.

Gef is released faster from nanoparticles prepared with 2 mg/mL chitosan than from those prepared with 4 mg/mL chitosan. These findings reveal that the release behavior of Gef is affected by the viscosity of the chitosan solution. When chitosan solution interacts with TPP, its increased viscosity constructs reasonably strong walls of nanoparticles. The TPP-chitosan matrix's high cross-linking density results in less swelling ability and, as a result, less Gef release [15].

It is evident from the bar graph in Fig. 1 that PS increased as chitosan concentration increased ($P < 0.05$). On the other hand, both EE (%) and %release decreased ($P < 0.05$).

Table 4
Model adequacy and validation for PS (nm), EE (%), and %release.

Formula	PS (nm)			EE (%)			%release		
	Actual	Predicted	Bias(%)	Actual	Predicted	Bias(%)	Actual	Predicted	Bias(%)
F1	106	114.58	8.10	71.3	70.94	−1.13	77.2	76.66	−0.69
F2	161	149.50	−7.14	58.2	57.89	−0.53	68.3	66.09	−3.24
F3	182	184.42	1.33	49.4	45.29	−8.31	57.9	55.51	−4.12
F4	145	131.33	−9.43	83.2	82.04	−1.39	85.8	87.01	1.41
F5	172	188	9.30	64.5	70.64	9.53	73.1	77.26	5.68
F6	246	244.67	−0.54	53.8	59.24	10.12	62.6	67.51	7.84
F7	158	148.08	−6.28	92.7	93.59	0.96	97.1	97.35	0.25
F8	201	226.50	12.69	87.1	83.39	−4.25	92.2	88.42	−4.10
F9	321	304.92	−5.01	75.6	73.19	−3.18	81.1	79.50	−1.98

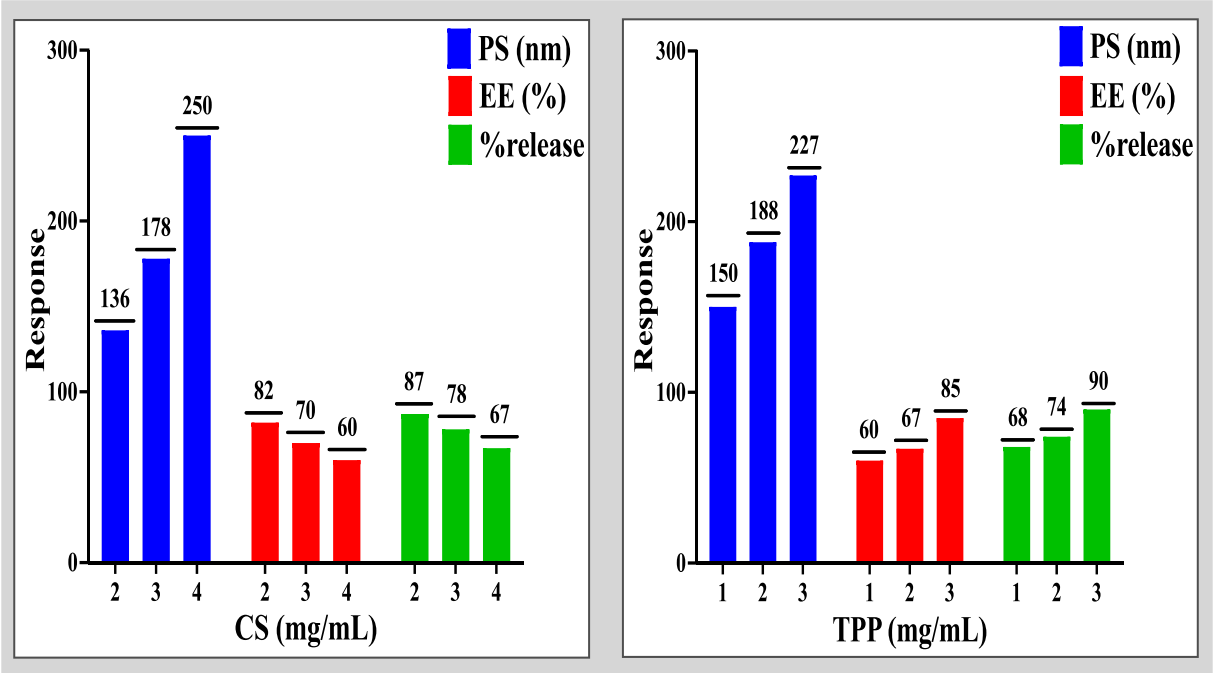

Fig. 1. Bar graph reveals that increasing chitosan increases the PS, decreases the %EE, and decreases the %release; and illustrates the upward trend of PS with increasing TPP concentration, the upward trend of %EE with increasing TPP, and the upward trend of %release with increasing TPP.

3.1.2. The impact of the TPP on PS, EE (%), and %release using a full factorial design

TPP had a significant effect on the PS, the EE (%), and the %release, according to the Pareto results in Table 2.

The PS rises as TPP concentrations rise. At TPP = 1 mg/mL, the mean PS (F1, F2, and F3) was 150 nm. TPP = 2 mg/mL (F4, F5, and F6) raised the mean PS to 188 nm, whereas TPP = 3 mg/mL (F7, F8, and F9) raised it to 227 nm.

Fig. 1 demonstrated the significant effect of TPP concentrations, which were directly related to PS enlargement [46]. At low TPP concentrations, the TPP was insufficient to completely cross-link the chitosan, and the preparation remained transparent. Increased TPP concentrations result in increased PS and a slightly opaque solution. Large-sized particles were obtained at higher TPP concentrations [43].

The EE (%) is proportional to the TPP concentration [15]. At TPP = 1 mg/mL, the mean EE (%) (F1, F2, and F3) was 60 %. TPP = 2 mg/mL (F4, F5, and F6) had a mean EE (%) of 67 %, but TPP = 3 mg/mL (F7, F8, and F9) had an 85 % mean EE (%) as depicted in Fig. 1. TPP had a positive effect on EE (%); as TPP increased, more Gef was trapped inside the nanoparticles [44].

In terms of %release, it was spiked with a higher concentration of TPP. The mean %release at TPP = 1 mg/mL (F1, F2, and F3) was 68 %.

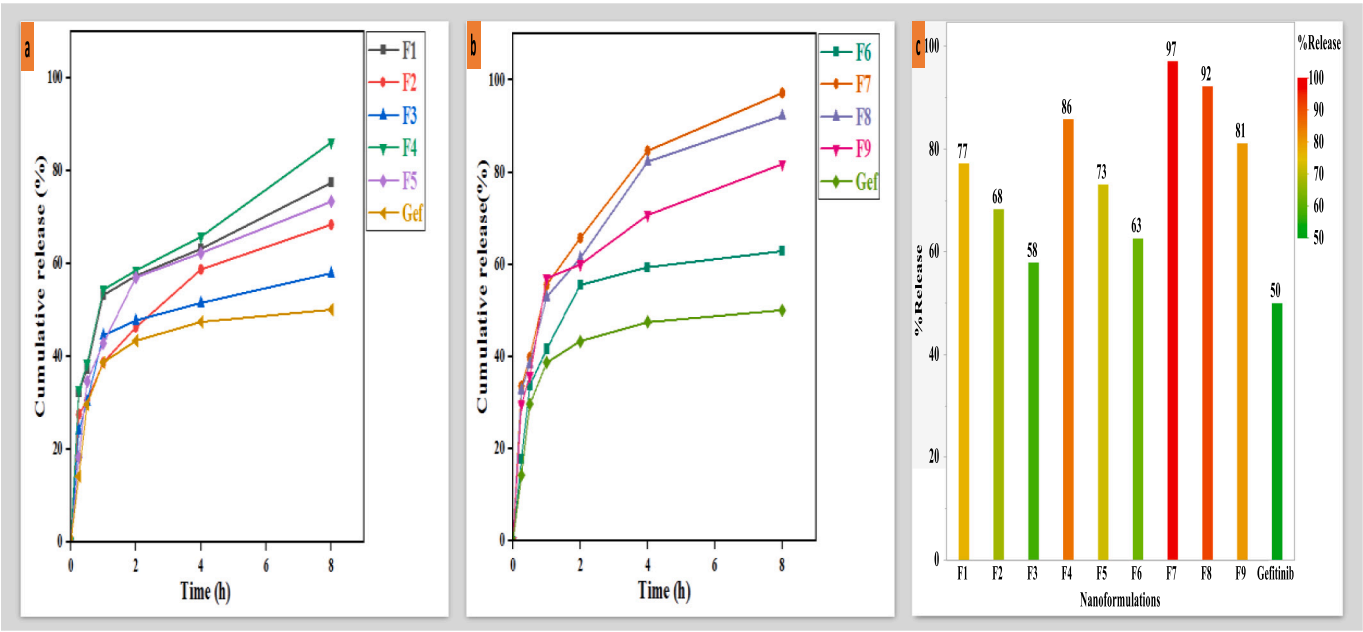

Fig. 2. *In vitro* drug release of the pure Gef and the 9 nanoformulations in FBS (pH of 5 at 37 °C and stirring rate of 100 rpm) using a dialysis bag. Data are the mean of 3 determinations.

The mean %release increased to 74 % at TPP = 2 mg/mL (F4, F5, and F6), and it peaked at 90 % at TPP = 3 mg/mL (F7, F8, and F9) as illustrated in Fig. 1.

Higher TPP levels increase the size of CSNPs, and as PS increases, the Gef-CSNPs erode faster in FBS. This causes the particles to lose their integrity quickly, resulting in rapid drug release [15]. It is evident from the bar graph in Fig. 1 that PS, EE (%), and %release increased as TPP concentration increased ($P < 0.05$).

3.2. Entrapment efficiency (%EE)

The EE (%) of Gef in the nine nanoformulations was calculated, and it ranges from 49 ± 0.52 to 93 ± 1.2 %. The highest EE (%) was found with F7 (93 ± 1.2 %) at 2 mg/mL chitosan and 3 mg/mL TPP. Bar graph showing the EE (%) of all the nanoformulations is available in supplementary materials.

3.3. In vitro release study

To predict and compare the bioavailability of pure Gef and Gef-CSNPs, an *in vitro* release study was performed. It is carried out in FBS at a pH 5 and a temperature of 37°C. The maximum %release was noted after 8 h. Therefore the %release of all the nanoformulations was recorded at this point of time.

Pure Gef has the lowest %release of 50 ± 1.4 % after 8 h, as shown in Fig. 2, whereas all nanoformulations have a much higher %release ranging from 58 ± 1.8 to 97 ± 0.6 % at the same period.

Six equations were used to model the release of the optimized Gef-CSNPs: zero-order, first-order, Higuchi, Korsmeyer-Peppas, Weibull, and Baker-Lonsdale. R^2 evaluated the best-fit model with the release data. Korsmeyer-Peppas was the best-fitting model, with the highest R^2 value of 0.9940.

Fitting the *in vitro* release data to the Korsmeyer-Peppas equation with the add-in DDsolver program (China Pharmaceutical University, Nanjing, China) yielded the drug release constant (K) and release exponent (n) of the optimized Gef-CSNPs. Microsoft Office Excel (Microsoft Corporation, Redmond, USA) was used as a DDSolver built-in module [47].

$$F = K \times t^n \quad (6)$$

where F is the fraction of drug release at time t.

K is the release constant.

n is the release exponent.

The release constant (K) is proportional to the kinetics of drug release from the nanocarrier. Indeed, a higher K value indicates faster drug release, whereas a lower K value indicates poor drug release from the nanocarrier. The optimized Gef-CSNPs had a K value of 53.07 h^{-1}, while pure Gef had a value of 33.09 h^{-1}. This discovery supports the much higher release of Gef-CSNPs when compared to pure Gef.

Using the release exponent (n), the Korsmeyer-Peppas model can also give detailed information about how drugs are released [47]. If $n < 0.45$, diffusion is the way the drug gets out. The n value of the optimized Gef-CSNPs was 0.305, which suggests that diffusion was in charge of the release process.

3.4. Fourier transform infrared (FTIR) spectroscopy

The FTIR spectra of pure Gef, chitosan, TPP, plain CSNPs, and Gef-CSNPs are shown in Fig. 3.

Gef demonstrates the following peaks: a sharp peak assigned to the secondary amine group at 3399.61 cm^{-1}, N—H bending vibration from 1532.66 to 1625.01 cm^{-1}, C—Cl peaks ranging from 545.08 to 774.65 cm^{-1}, C—F peaks ranging from 1028.72 to 1397.68 cm^{-1}, and C—O peaks at 1110.18 cm^{-1}. The -C=N peak is at 1902.9 cm-1. The aromatic region (900–650 cm^{-1}) contain peaks ranging from 654.56 to 911.75 cm^{-1}, with a = C-H peak at 3100.14 cm^{-1} [48].

Chitosan spectrum shows: The forked peak at 2923 and 2853 cm^{-1} is attributed to the primary amine (−NH2), the broad band at 3401 cm^{-1} indicates the presence of -OH stretching vibration, and the symmetric stretch of C-O-C is located at 1110 cm^{-1}. The C—N— stretching is responsible for the strong peak at 1429 cm^{-1}. The IR spectrum of chitosan exhibits the -N-H bending peak at 1578 cm^{-1} and the band for amide at 1625 cm^{-1}. Additionally observed is the $-NH_2$ bending vibration at 1625 cm^{-1} [26,49].

Because of the strong ionic interaction between the amino groups of chitosan and the phosphate group of TPP, the -OH stretching vibration peak shifts from 3401 to 3417 cm^{-1} and the -NH2 bending vibration shifts from 1625 to 1731 cm^{-1} in Gef-CSNPs.

Peaks of Gef in the aromatic region vanished, indicating Gef encapsulation, and the high similarity between plain CSNPs and Gef-CSNPs indicates that Gef encapsulation inside CSNPs is highly efficient. Due to its encapsulation, Gef's characteristic peaks are missing.

3.5. Particle size (PS) and morphology analysis by DLS, TEM, and SEM

PS analysis is used to study dispersion and aggregation. Large particles are more likely to aggregate than smaller particles, resulting in deposition. PDI is a measurement of nanoparticle homogeneity. A PDI value close to zero indicates homogeneous nanoparticle dispersion, while a PDI value >0.5 indicates a high degree of heterogeneity. The DLS results indicate that the nine batches of Gef-CSNPs have a PS range

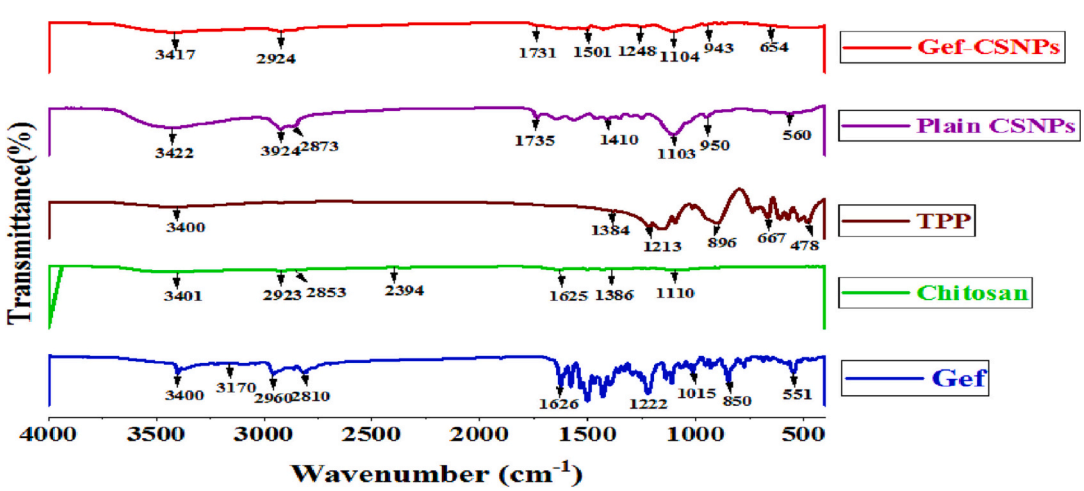

Fig. 3. FTIR spectra of pure Gef, chitosan, TPP, plain CSNPs, and the optimized Gef-CSNPs.

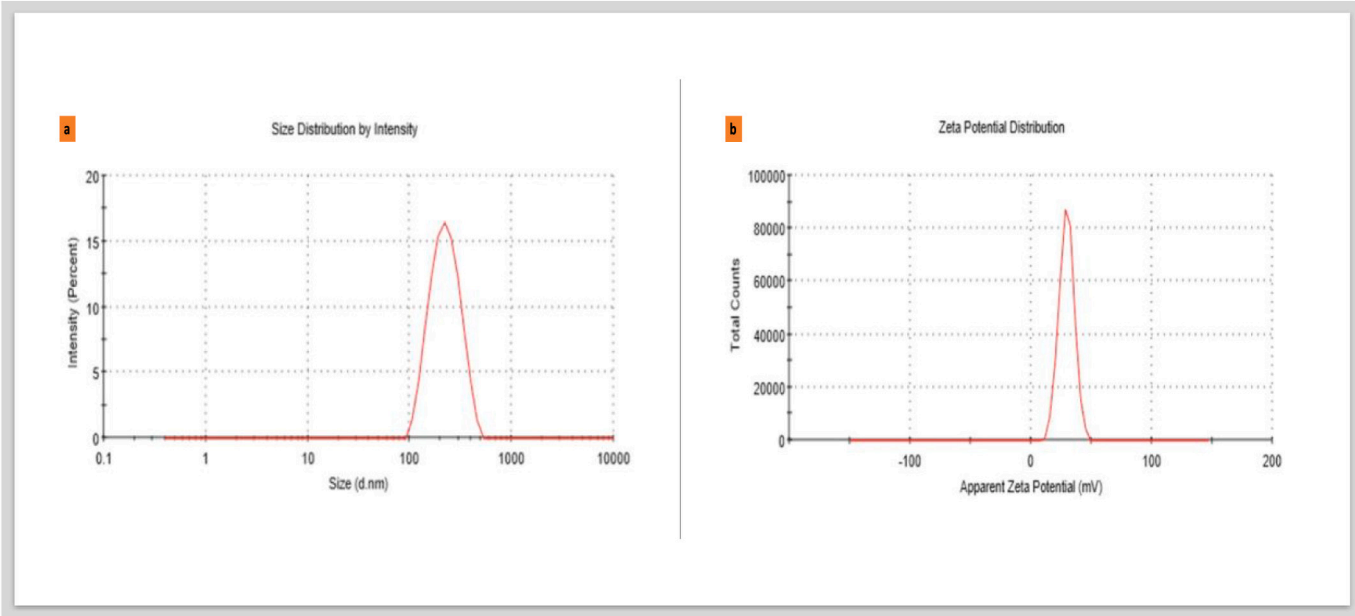

Fig. 4. Particle size distribution (a) and zeta potential of the optimized Gef-CSNPs (b) by zetasizer (size $= 158\,\mathrm{nm}$, ZP $= +30.2\,\mathrm{mV}$).

of 106–321 nm and a PDI range of 0.11 to 0.38 (PDI <0.5), indicating uniformity. As a result, chitosan can entrap Gef at the nanoscale. The PS determined by DLS in Fig. 4 was higher than that determined by TEM because DLS analysis is performed in solution and the particles are susceptible to agglomeration, thereby increasing PS, whereas TEM measurement represents a dried layer of nanoparticles on a TEM grid and there is no chance of aggregation [50–52]. Zeta potential (ZP) depicted in Fig. 4 is regarded as a crucial marker of colloidal dispersions' physical stability. The tendency to stabilize colloidal preparations will increase with increasing *ZP* values, whether positive or negative. In formulations including charged particles with a ZP value of >20, the aggregation of particles is typically prevented. It takes this much ZP to keep the electrostatic attraction between the similarly charged particles strong enough to prevent aggregation. The optimized Gef-CSNPs had a zeta potential of +30.2 mV [53].

Fig. 5 depicts the TEM and SEM morphology of plain CSNPs and Gef-CSNPs. TEM image of Gef-CSNPs displays a narrow PS distribution, ranging from 79 nm to 128.4 nm with an average size of 101.8 nm, which was smaller than the DLS measurement, as expected.

SEM images of Gef-CSNPs and plain CSNPs revealed smooth and spherical particles with aggregation [51,52]. The SEM micrographs revealed no discernible differences in the surface morphology of plain CSNPs and Gef-CSNPs, confirming Gef encapsulation within the CSNPs.

The optimized Gef-CSNPs' particle size distribution histogram (supplementary materials) was derived from SEM images using imageJ software, based on 14 points selection. It showed a narrow range of particle sizes. The size of the particles ranges from 106 to 240 nm, with an average diameter of 166 ± 44.9 nm. An average diameter of 163 ± 24.7 nm, and a PS range of 114–194 nm are all characteristics of plain CSNPs. It is noticed that loading Gef inside CSNPs had led to minimal

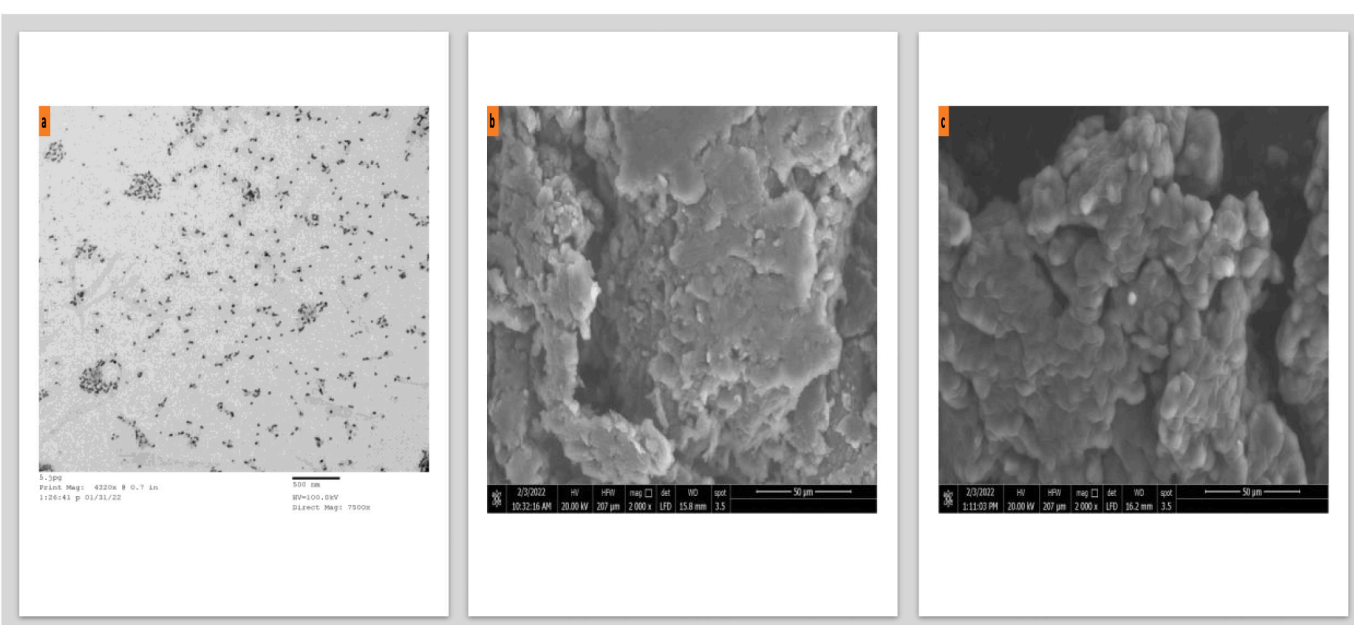

Fig. 5. TEM micrograph of the optimized Gef-CSNPs (a), SEM micrograph of the optimized Gef-CSNPs (b), and plain CSNPs (c).

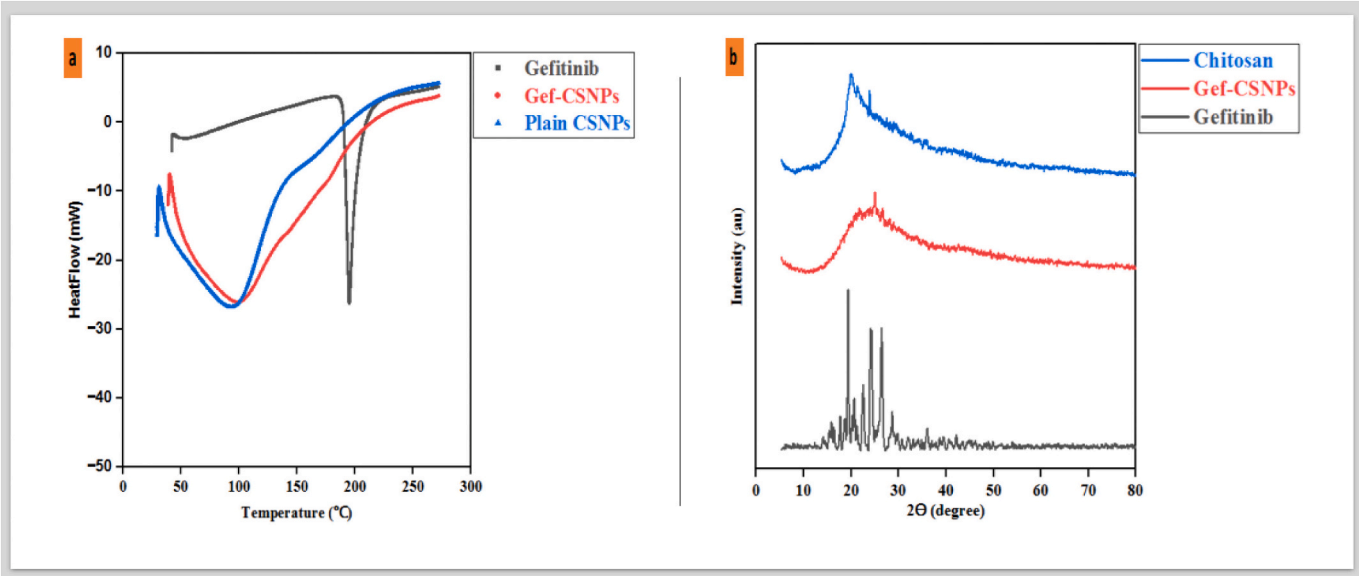

Fig. 6. DSC thermograms of pure Gef, the optimized Gef-CSNPs, and plain CSNPs (a) and XRD spectra of chitosan, the optimized Gef-CSNPs, and pure Gef (b).

increase in PS.

3.6. Differential scanning calorimetry (DSC)

Fig. 6(a) depicts the thermograms of Gef, plain CSNPs, and Gef-CSNPs. The DSC thermogram of pure Gef reveals a sharp endothermic peak at 195.067°*C* [54], with a melting temperature range of 184.54 to 236.90°*C* and a heat of fusion (ΔH) of 166.597 (J/g), indicating that Gef is crystalline.

The plain CSNPs demonstrates an endothermic melting temperature of 92.864°C with a melting temperature range of 31.21 to 148.26°C and a heat of melting (ΔH) of 290.705 (J/g). The degradation temperature value of plain CSNPs was attributed to the second endothermic peak observed at 165.051°C with the heat of melting equal to 0.522 (J/g).

Gef-CSNPs have an endothermic peak at 94.769°C, a melting temperature range of 39.99 to 139.35°C, and an enthalpy (ΔH) of 201.784 (J/g). The degradation temperature of Gef-CSNPs was attributed to the second endothermic peak observed at 177.496°C with the heat of melting equal to 0.363 (J/g). The characteristic Gef peak faded away from the Gef-CSNPs, indicating that Gef had completely dispersed in the CSNPs [15,42,55].

3.7. X-ray diffraction (XRD)

Gef's diffractogram depicted in Fig. 6(b) revealed distinct sharp diffraction peaks at 2θ values of 19.348°, 20.655°, 22.523°, 24.230°, 26.389°, 28.616°, 35.978°, 42.124°, and 49.903°. Previous studies reported similar peaks, which were attributed to Gef crystallinity. Diffraction patterns of chitosan were observed at 2θ values of 19.376° and 23.872° [56,57].

Regarding CSNPs, the crystalline peak at 2θ = 19.376° vanished and amorphous peaks were observed at 2θ = 25°–27°.

The sharp crystalline Gef peaks have vanished from the diffractogram of Gef-CSNPs, indicating that Gef was dispersed in the CSNPs, while the fundamental peaks of chitosan have been preserved and peak intensity has been reduced. The diminishing of the peaks of Gef-CSNPs is highly evident and is linked to the presence of Gef in the amorphous phase.

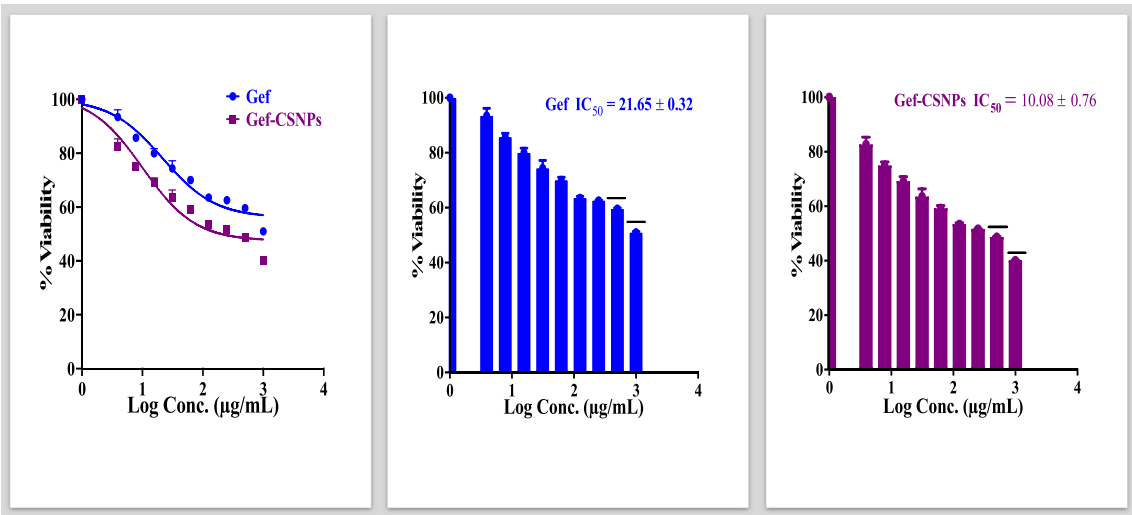

Fig. 7. Cytotoxicity profile of Gef and Gef-CSNPs against A-549 lung cancer cell line, with IC_{50} values of (21.65 ± 0.32 μg/mL) and (10.08 ± 0.76 μg/mL), respectively, illustrates the powerful antiproliferative effect of the optimized Gef-CSNPs over pure Gef.

3.8. In vitro cytotoxicity (cell viability assay)

The MTT assay was used to investigate the effect of Gef and Gef-CSNPs on the viability of A549 cells. All experiments were carried out in triplicate. The IC_{50} value was calculated in GraphPad Prism software version 9.3.1 using the non-linear regression fit method. The following formula was used to calculate cell viability:

$$\%\text{Cell viability} = \frac{(\text{OD of sample} - \text{OD of blank})}{(\text{OD of control} - \text{OD of blank})} \times 100 \quad (7)$$

As illustrated in Fig. 7, Gef has high cytotoxicity against A549 cells (IC_{50}=21.65 ± 0.32 μg/mL) [58], but the cytotoxic effect of Gef-CSNPs was more pronounced, with an IC_{50} value that was (10.08 ± 0.76 μg/mL). As a result, it is clear that Gef-CSNPs inhibited A549 cell growth more effectively than pure Gef. These findings revealed that more Gef-CSNPs could be internalized into A549 cells, resulting in increased cytotoxicity due to Gef accumulation in the cells. These findings show that chitosan played an important role as a nanocarrier for Gef delivery inside the tumor.

3.9. Apoptosis assay

Flow cytometry is used to investigate the mechanisms of cell death. The flow cytometry scatter plot is divided into four quadrants, with the first (Q3) representing viable cells that are not attached to Annexin-*V* or PI (Annexin-V negative and PI negative or double negative). The second quadrant (Q4) contains early apoptotic cells that bind only to Annexin-V and not PI (Annexin-V positive and PI negative). The third quadrant (Q2) represents late apoptotic cells, which are Annexin-V and PI positive (double-positive). Aside from apoptosis, some cells may undergo necrosis (Q1) [59]. The cells are PI-positive and Annexin-V-negative. Using this clever technique to investigate the mechanism of A549 cell death caused by Gef [60] and Gef-CSNPs. In A549 cells exposed to Gef and Gef-CSNPs, the assay confirmed apoptosis.

Figs. 8 and 9 display that A549 cells exposed to doxorubicin (positive control) showed no significant apoptosis (3.07 ± 0.65 %); however, Gef and Gef-CSNPs exposed A549 cells induced higher apoptotic cell populations (29.38 ± 1.11 %) and (64.82 ± 1.25 %), respectively. These findings support the MTT assay findings of increased cytotoxicity caused by Gef-CSNPs.

3.10. In vitro cellular uptake

Cancer cells have limited uptake of chemotherapeutics, which contributes to multidrug resistance (MDR). This MDR is responsible for >90 % of deaths in cancer patients receiving traditional chemotherapeutics; thus, it is critical to design a system that can avoid this MDR induced by cancer cells. This appears to be a capability of nanoparticles.

This was confirmed by using HPLC to compare the uptake of pure Gef and Gef-CSNPs inside A549 cells [61,62]. First, a six-point calibration curve for Gef was performed over a concentration range of 0.1–1 μg/mL (0.1, 0.2, 0.4, 0.6, 0.8, and 1 μg/mL). The calculations were based on a plot of peak area *versus* concentration as shown in Fig. 10(a). The results shown in Fig. 10(b) show that uptake from Gef-CSNPs is nearly double (3.286 ± 0.12 μg/mL) that of pure Gef (1.777 ± 0.1 μg/mL) [63]. Because smaller PS is easily uptaken by cancer cells, these findings highlight the role of nanoparticles in fighting lung cancer due to their higher degree of internalization inside tumor cells [64].

4. Conclusion

Cancer kills someone every five seconds around the world. The National Cancer Institute reported 135,720 lung cancer deaths in the United States in 2020. These findings have served as a wake-up call to governments, healthcare systems, and researchers. To alleviate cancer patients' suffering, researchers are racing to develop an effective and safe therapy for lung cancer. Traditional chemotherapeutics' failure has shifted researchers' attention to nanoparticles. We used chitosan nanoparticles to deliver Gef to cancer cells, which prevented it from being misplaced. As a result, side effects are reduced, solubility and bioavailability are improved, therapeutic control is improved, cellular uptake into A549 cells is increased, and cytotoxicity is increased, as evidenced by a much higher apoptotic effect and cell growth inhibition. These findings show that the efficacy of Gef-CSNPs is beyond any

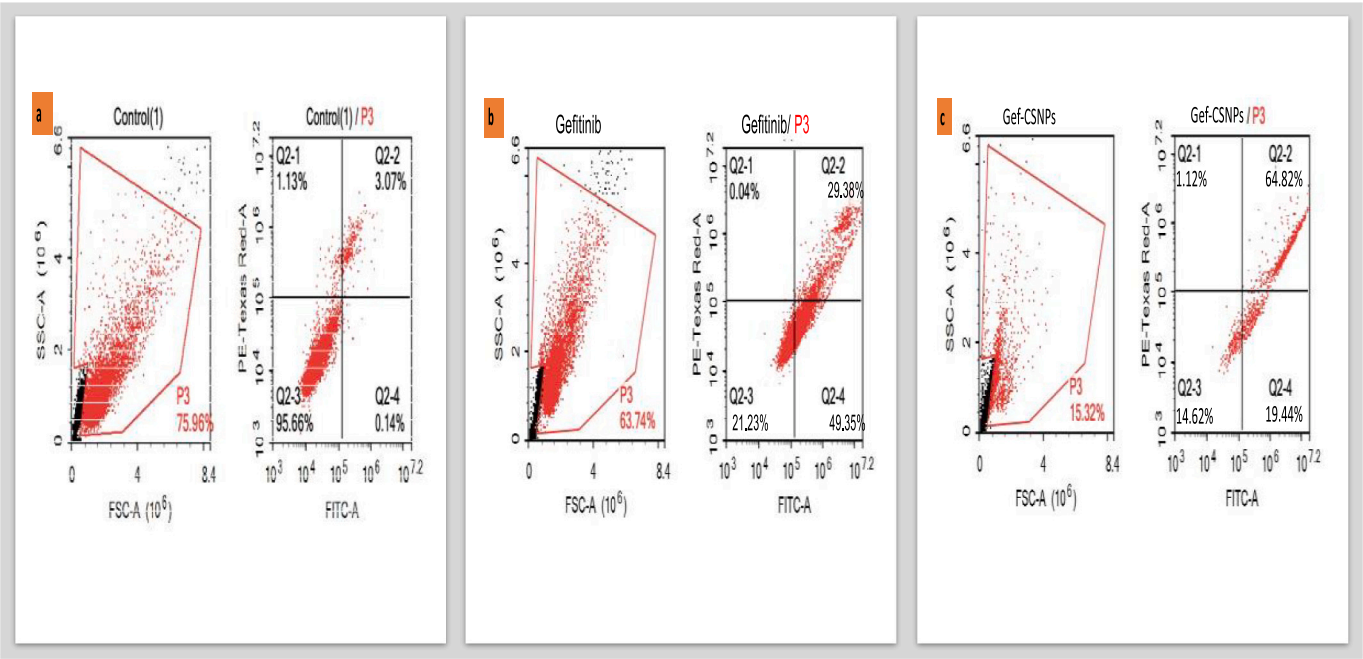

Fig. 8. A scatter plot depicting apoptosis of A549 cell populations exposed to doxorubicin (control), revealing %apoptosis of 3.07 ± 0.65 % (a), exposed to the pure Gef, revealing %apoptosis of 29.38 ± 1.11 % (b), and exposed to Gef-CSNPs, revealing %apoptosis of 64.82 ± 1.25 % (c). [necrotic cells(Q1), late apoptotic cells (Q2), healthy viable cells (Q3), and early apoptotic cells(Q4)].

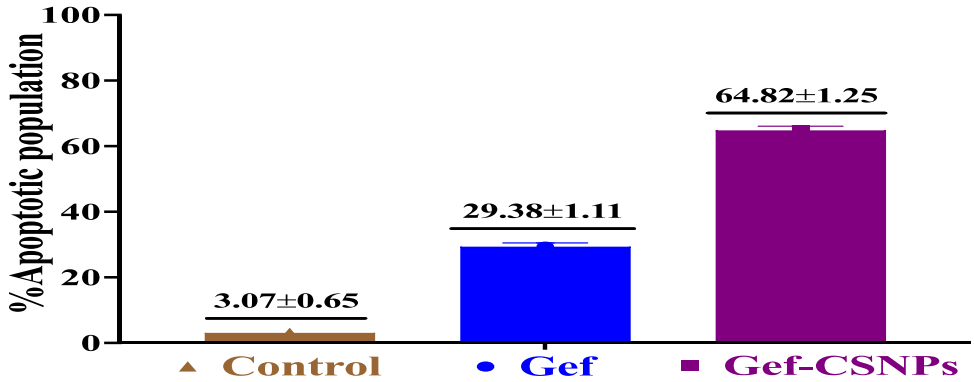

Fig. 9. Gef and Gef-CSNPs elicited apoptotic population of (29.38 ± 1.11 %) and (64.82 ± 1.25 %), respectively.

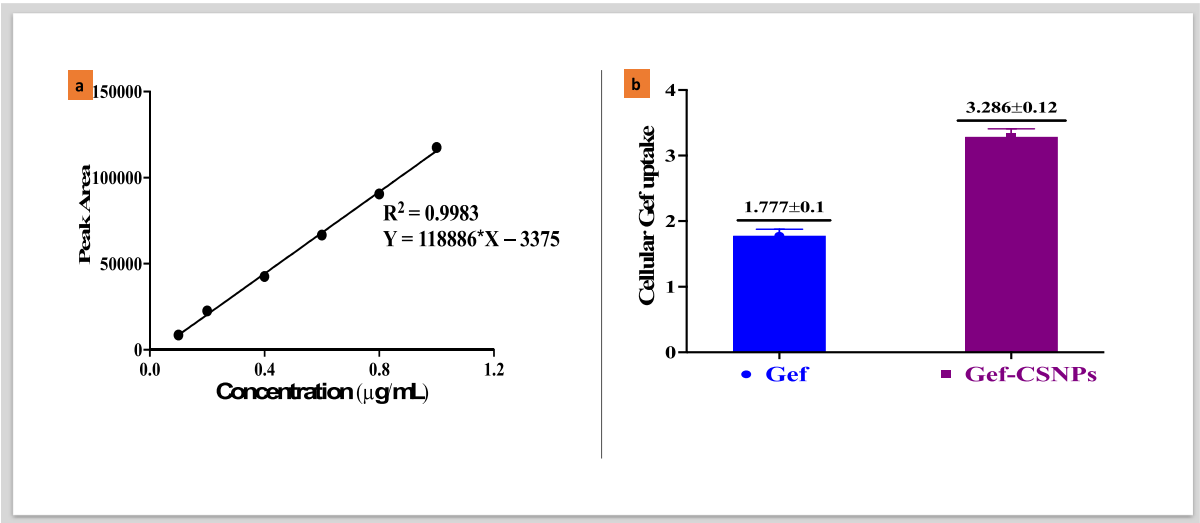

Fig. 10. HPLC calibration curve of the pure Gef (a), cellular uptake of Gef and Gef-CSNPs into A549 human lung cancer cells demonstrates that Gef-CSNPs have a significantly higher uptake due to their nano-size (b).

reasonable doubt, and a lot of hope is riding on nanoparticles derived from nature. Wrestling the cancer dragon appears to be a difficult task, but chitosan nanoparticles provide a window of opportunity. Despite these spectacular results, the main limitation of this work is the lack of *in vivo* evaluation of the developed system, but this opens the door for additional research to confirm *in vivo* efficacy.

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CRediT authorship contribution statement

Haitham Amin; Conceptualization, methodology, software, statistical analysis, data interpretation, and writing the paper.
Mohammed A. Amin; data interpretation, review, and editing.
Shaaban K. Osman; Supervision.
Ahmed M. Mohammed; Supervision.
Gamal Zayed; Conceptualization, supervision, review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2023.125638.

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